

Use Case “Add/Update Product (Product Manager)”

1. Use case code

UC001

2. Brief Description

This use case describes how a Product Manager adds a new product or updates an existing product in AIMS. The system enforces product-type requirements (Book/Newspaper/CD/DVD), validates the price band (current price must be 30%–150% of the original value), and stores a history of product additions and edits. If an input is invalid, the system notifies the Product Manager.

3. Actors

3.1. Product Manager

4. Preconditions

- The Product Manager is logged in with the appropriate role to access product-maintenance features.
- For Update: the target product already exists in the catalog.

5. Basic Flow of Events (Success)

1. Product Manager requests to Add a new product or Update an existing product.
2. Software displays the product form with a product type selector (Book, Newspaper, CD, DVD).
3. Product Manager enters product data, including title/category (AIMS supports search by product title or category), original value, current price, and type-specific fields as required.
4. Software validates:

- Price policy: current price is within 30%–150% of the original value.
- Type-specific required fields are present (per product type list below).
- Date formats are valid where applicable (e.g., publication date). If invalid, the system notifies the Product Manager.

5. Software performs the requested action:

- Add: creates a new product record.
- Update: applies the submitted changes to the existing record.

6. Software stores a history entry for the addition or edit.

7. Software confirms success and shows the saved product summary.

6. Alternative Flows

Table 2 — Alternative flows of events for UC “Add/Update Product”

No	Location	Condition	Action	Resume location
1	At Step 4	Price violates the 30%–150% band	Software notifies invalid price; submission rejected	Step 2
2	At Step 4	Missing type-specific required info	Software notifies invalid/insufficient info	Step 2
3	At Step 4	Invalid date format (e.g., publication date)	Software notifies invalid format	Step 2
4	At Step 3 (Update)	Product Manager adjusts stock quantity manually	Software requires reason for stock change and records it	Step 4

7. Input data

Table A — Input data for Add/Update Product

A.1 Common fields (all products)

No	Data fields	Description	Mandatory	Valid condition / Notes
1	Title	Title of products	Yes	Title exists
2	Category / Product type	(Book, Newspaper, CD, DVD)	Yes	One of the supported product types
3	Original value	Implicitly inferred from the requirements	Yes	> 0; used for price-band validation (see A.6)
4	Current price	The current price after adjustments from the original price	Yes	Must be within 30%–150% of original value
5	Stock quantity (if updating quantity)		Conditional (Update)	Manual adjustments must record a reason

A.2 Book-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Author(s)	Book authors	Yes	Non-empty	“Robert C. Martin”
2	Cover type	Paperback or hardcover	Yes	One of {Paperback, Hardcover}	Hardcover
3	Publisher	Publishing house	Yes	Non-empty	Pearson
4	Publication date	Date of publication	Yes	Valid date format (the system notifies if date format is incorrect)	30/12/2018

A.3 Newspaper-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
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1	Editor-in-chief	Lead editor	Yes	Non-empty	Nguyen Van A
2	Publisher	Publishing organization	Yes	Non-empty	Vietnam News Agency
3	Publication date	Date of issue	Yes	Valid date format (the system notifies if date format is incorrect)	05/10/2025
4	Issue number	Issue identifier	No	—	342
5	Publication frequency	Daily/Weekly/etc.	No	—	Weekly
6	ISSN	International Standard Serial Number	No	—	0868-1234
7	Language	Publication language	No	—	Vietnamese
8	Sections	e.g., politics, business, sports, culture	No	—	Politics; Business

A.4 CD-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Artists	Performer(s)	Yes	Non-empty	“Adele”
2	Record label	Label/publisher	Yes	Non-empty	XL Recordings
3	Tracks list	List of tracks	Yes	Each track has title and length	“Track 01 – Hello – 4:55”
4	Genre	Musical genre	Yes	Non-empty	Pop
5	Release date	Date of release	No	—	20/11/2015

A.5 DVD-specific (required)

No	Data fields	Description	Mandatory	Valid condition	Example
1	Disc type	Blu-ray or HD-DVD	Yes	One of {Blu-ray, HD-DVD}	Blu-ray
2	Director	Film director	Yes	Non-empty	Christopher Nolan
3	Runtime	Total running time	Yes	Positive duration	152 min
4	Studio	Production studio	Yes	Non-empty	Warner Bros.
5	Language	Audio language(s)	Yes	Non-empty	English
6	Subtitles	Subtitle language(s)	Yes	Non-empty	English; Vietnamese
7	Release date	Date of release	No	—	16/07/2010
8	Genre	Film genre	No	—	Action

Validation note: When product data **does not comply with rules** or **date format is incorrect**, the software **notifies** the Product Manager.

8. Output data

Table B — Output data after successful Add/Update

No	Data fields	Description
1	Product title & type	The saved product and its type (Book/Newspaper/CD/DVD)
2	Original value & current price	Confirmed values satisfying the 30%–150% band
3	Type-specific attributes	The saved attributes according to the product type (see A.2–A.5)

4	History entry recorded	Addition/edit stored in history
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9. Postconditions

- Add: A new product exists in the catalog; a history entry is recorded.
- Update: The product reflects submitted changes; a history entry is recorded.
- Saved current price respects the 30%–150% policy; required type-specific attributes are present.